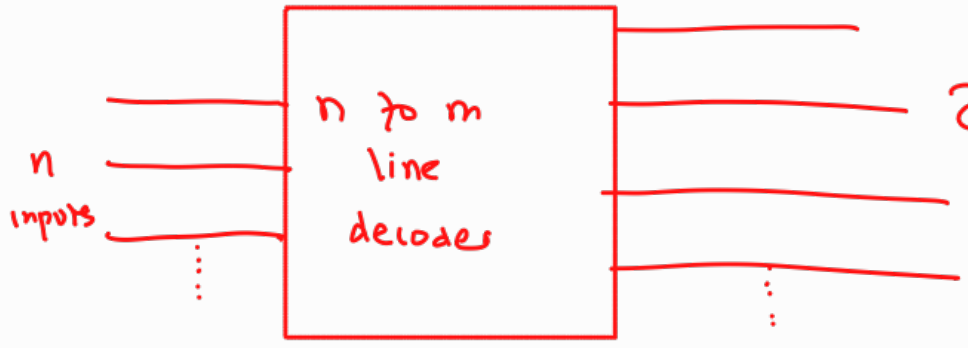
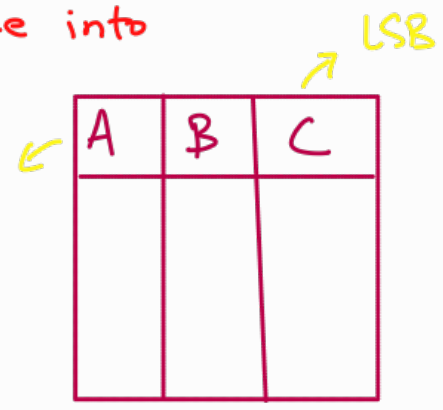
Decoding is the conversion of n-bit input code into an m-bit output code

generate diagram / Block diagram





(generates minterms in output) → gives 1 for corresponding combination and 0 for all of the rest.

e.g of decoders

→ 1 to 2 line
2 to 4 line
3 to 8 line
4 to 16 line

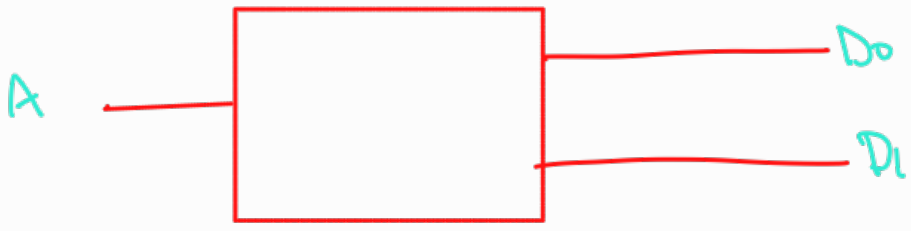
Decoder	Application

→ BCD to Seven Segment display.

→ To in memory allocation.

1 to 2 line decoder

n = 1 → inputs m = 2^1 = 2 outputs | minterms



D_i → i is decimal equivalent

(How to solve)

e.g Input single value is 10 in Binary. Now convert it to decimal 10 → 2. Hence it will go to D_2 output meaning High output will be at D_2 only while rest will be zero

1 to 2 line T.T

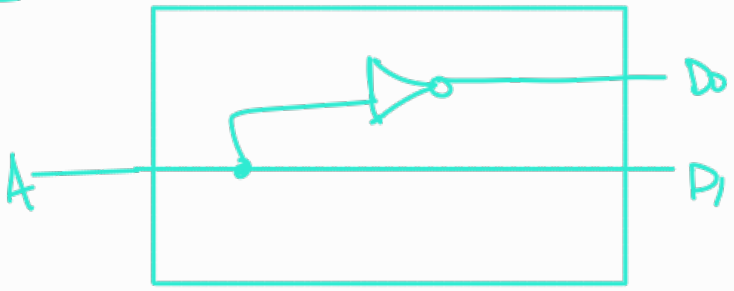
Input A	output D_0	D_1
0	1	0
1	0	1

→ convert to decimal

Making equations for minterm

$D_0 = \bar{A}$ > Inputs

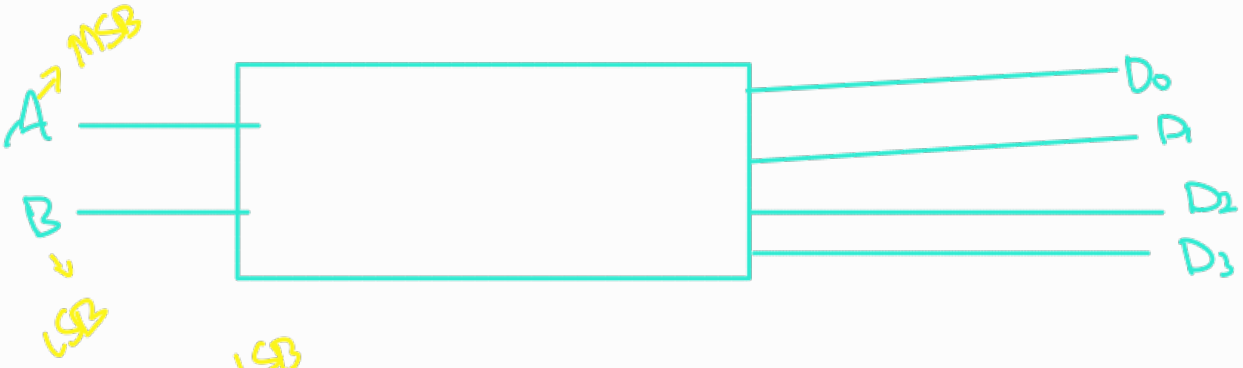
$D_1 = A$



Circuit diagram ←

2 to 4 line decoder

$m = 2^n = 2^2 = 4$ outputs



A (MSB)	B (LSB)	D_0	D_1	D_2	D_3
0	0	1	0	0	0
0	1	0	1	0	0
1	0	0	0	1	0
1	1	0	0	0	1

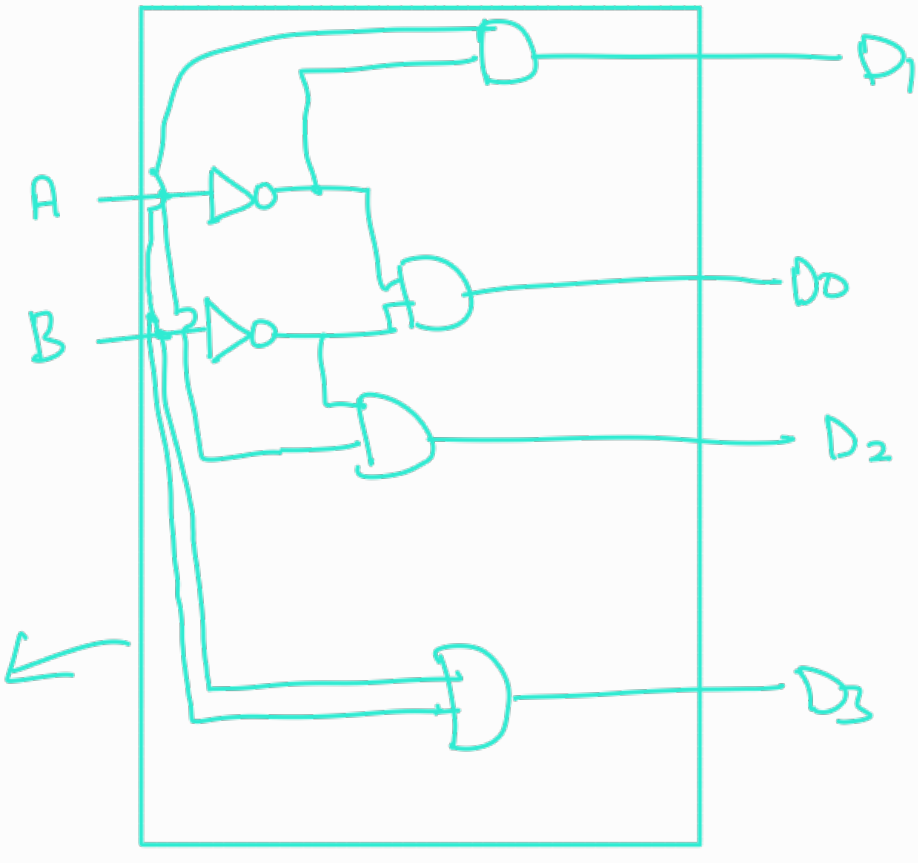
Minterms $(\bar{1} \cdot \bar{1}) \rightarrow 1$

$D_0 = \bar{A}\bar{B}$

$D_1 = \bar{A}B$

$D_2 = A\bar{B}$

$D_3 = AB$



circuit Diagram

3 to 8 Line decoder $2^3 = 8$ outputs

note e.g which output signal will be ON on input $(110)_2$

↓

hence converting it decimal is 6

m_6 / D_6

4 to 16 line decoder → $2^4 = 16$ outputs

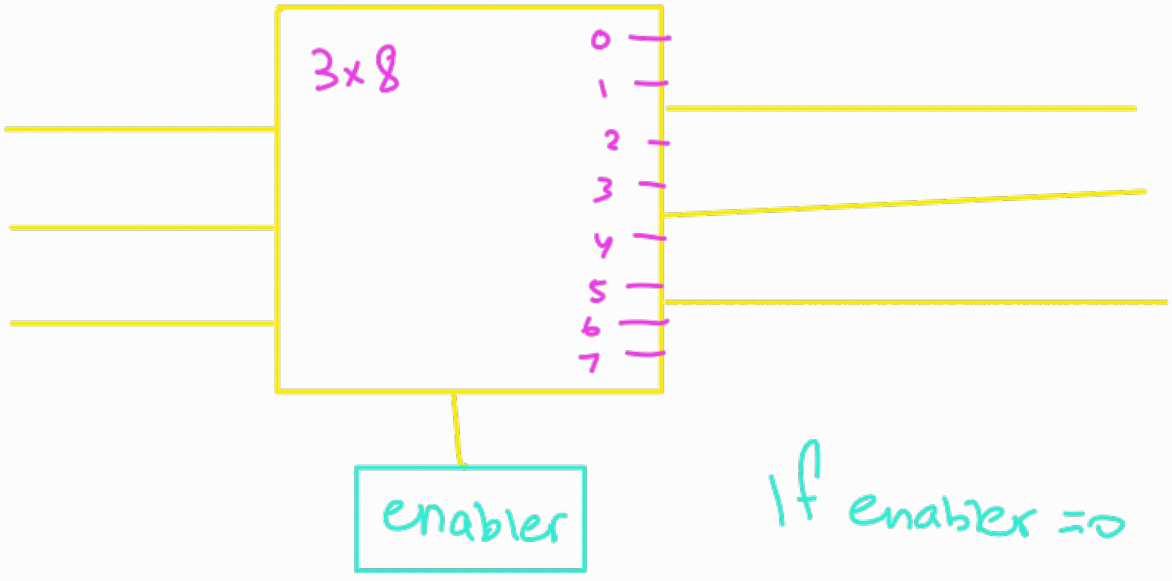
Decoder with enable input

↓

Similar to Power Supply of a decoder

Benifits of E.I → helps in making Larger order decoder from Smaller order decoder.

→ diff from decoder inputs



if enabler = 1

decoder = ON

↳ depending on input

the corresponding output line is 1 while all other are 0

If enabler = 0

decoder = off ↓

all outputs lines are 0

(Decoder with enables example)

Input

2x4 Decoder $2^2 = 4$ outputs

A	B	En	D_0	D_1	D_2	D_3
X	X	0	0	0	0	0 ← off
0	0	1	1	0	0	0
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	1

We attach En with And gates b/w En and Ans of $D_0, \dots D_3$.

Making equations so that we can make the logic circuit along with enabler input

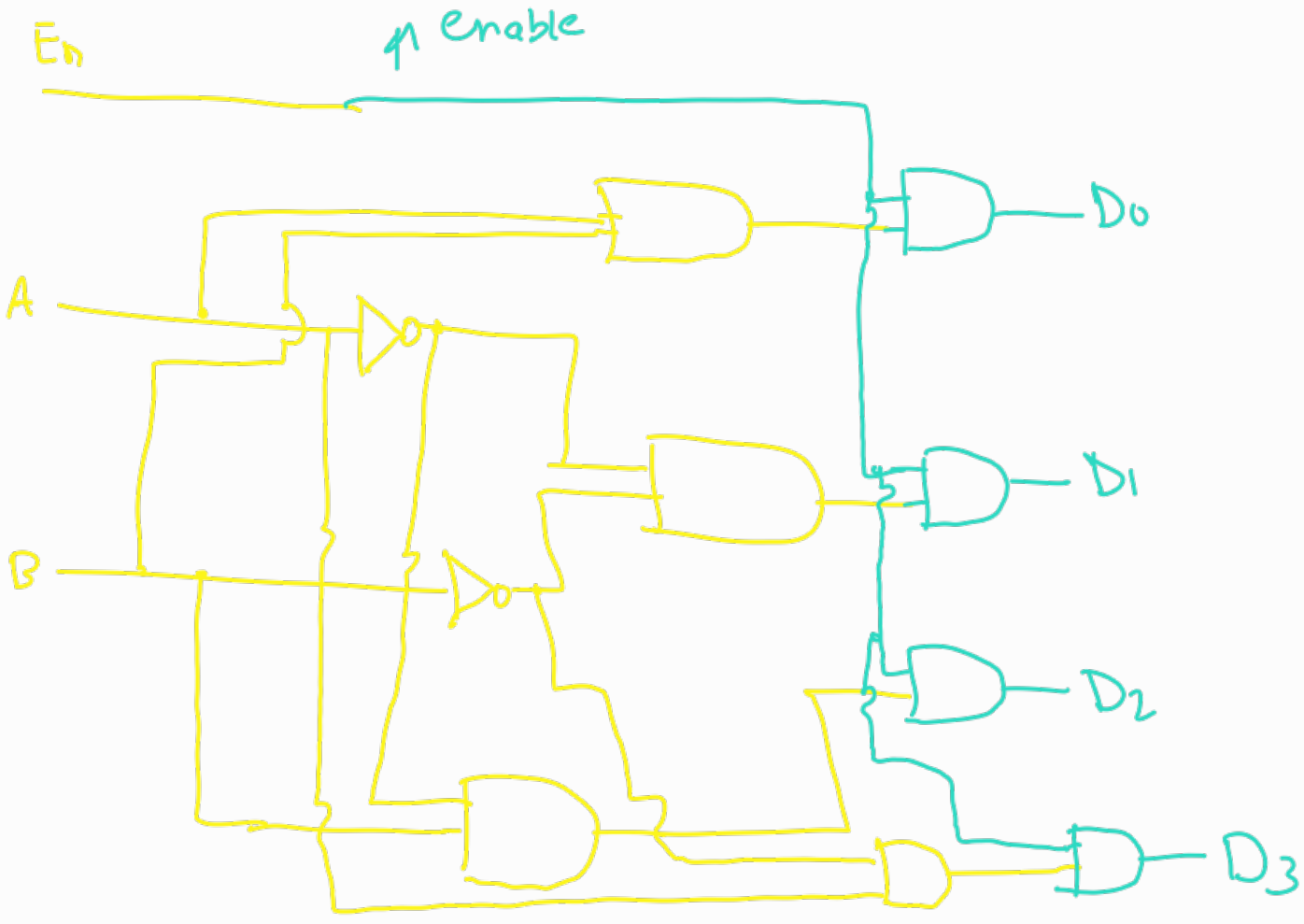
Equation

$(\bar{A}\bar{B} + \bar{A}B + A\bar{B} + AB)$

Equation

$(\bar{A}\bar{B}, \bar{A}B, A\bar{B}, AB)$

$D_0 \quad D_1 \quad D_2 \quad D_3$



Implementing function using decoder

(Look at where output is 1)

→ Full bit Binary adder:. → Add 3 bits together

x	y	z	C	S
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	1	0
1	0	0	0	1
1	0	1	1	0
1	1	0	1	0
1	1	1	1	1

1+1 → 0 carry 1

1+1+1 → 1 carry 1

Question:: 3x8

Using decoder and OR gate for implementation.

Two outputs will be generated.

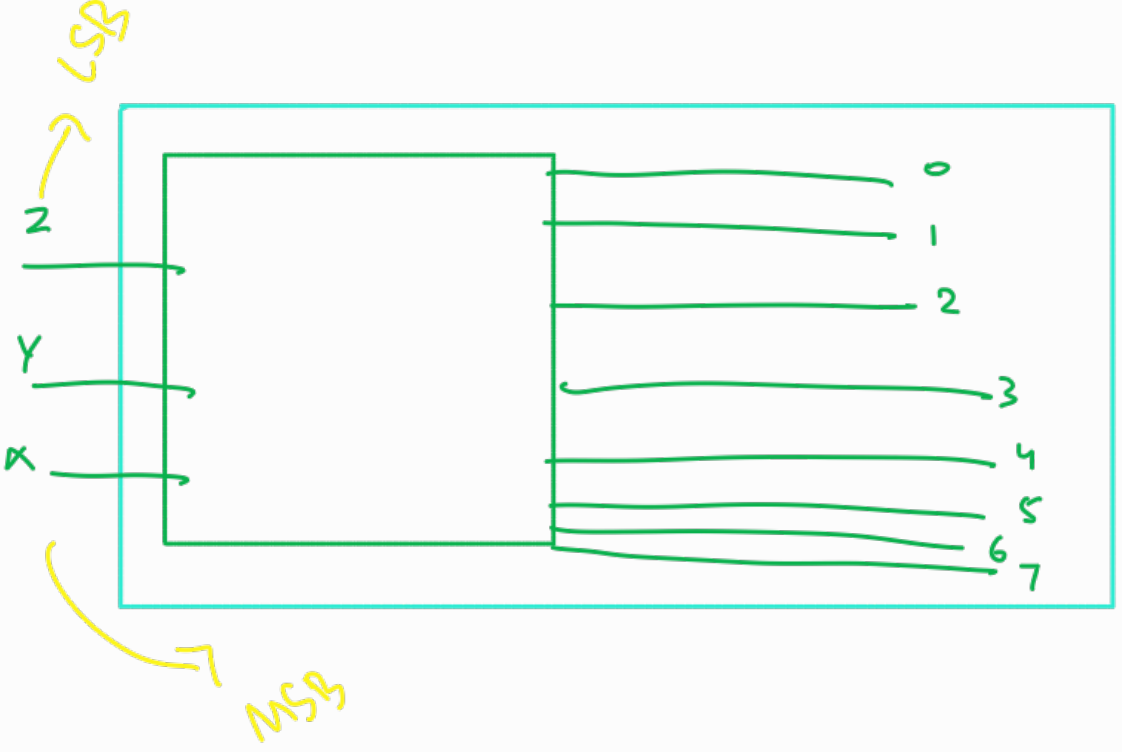
x	y	z	C	S	
0	0	0	0	0	m_0
0	0	1	0	1	m_1
0	1	0	0	1	m_2
0	1	1	1	0	m_3
1	0	0	0	1	m_4
1	0	1	1	0	m_5
1	1	0	1	0	m_6
1	1	1	1	1	m_7

For C

$\bar{x}yz$, $x\bar{y}z$, $xy\bar{z}$, xyz $\rightarrow$ $\Sigma m(3,5,6,7)$

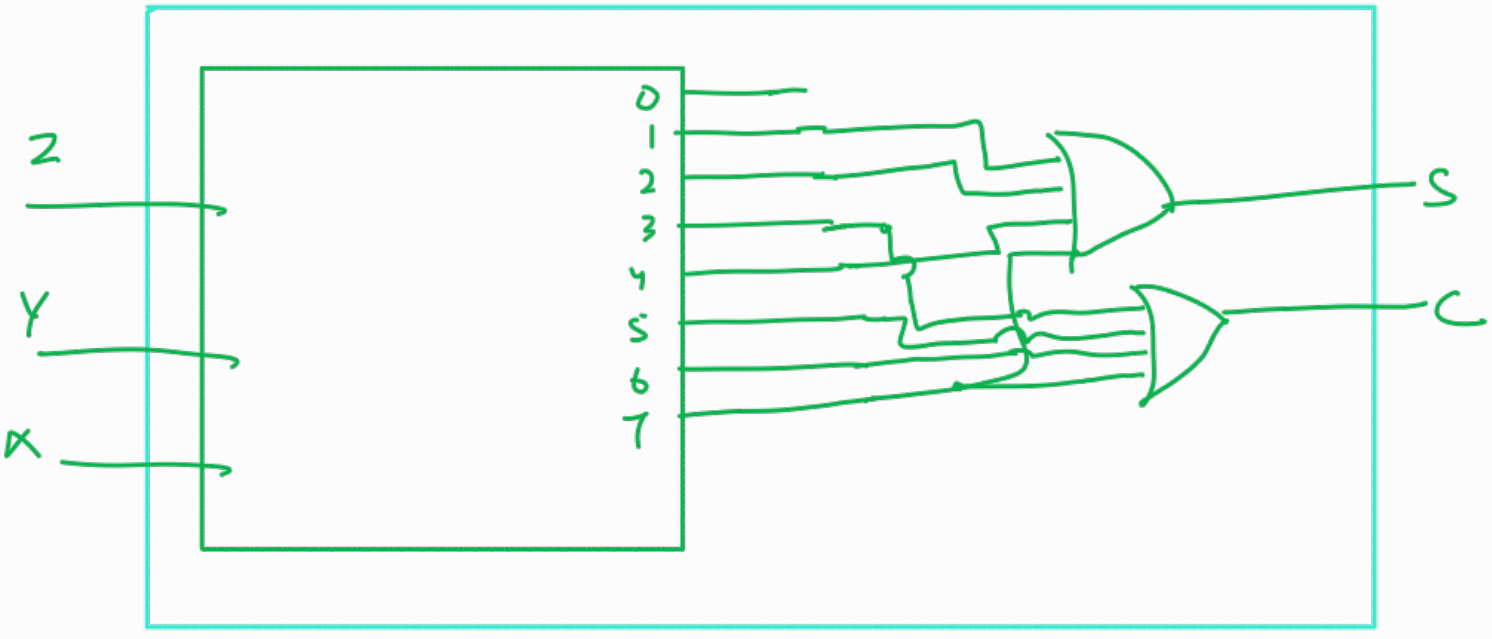
For S

$\bar{x}\bar{y}z$, $\bar{x}y\bar{z}$, $x\bar{y}\bar{z}$, $xy\bar{z}$, xyz $\rightarrow$ $\Sigma m(1,2,4,7)$



We can use any time of decoder as it is not mentioned

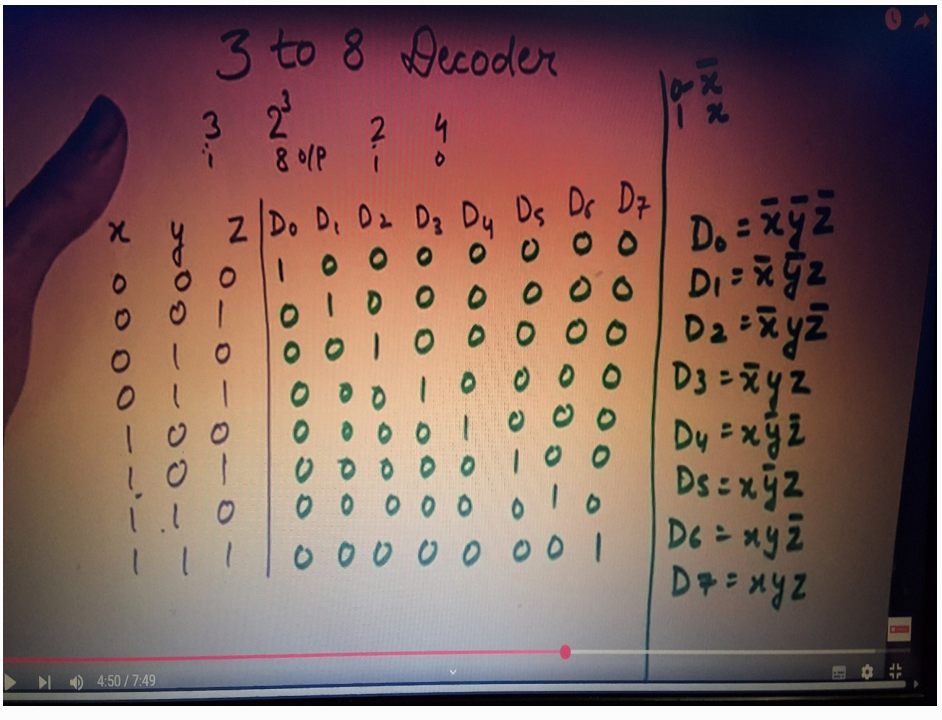
Using 3 x 8 as in T.T we have 3 inputs and 8 output



To Implement a function we need m OR gates

or

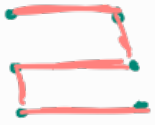
Same no. of OR gates as total output in T.T.



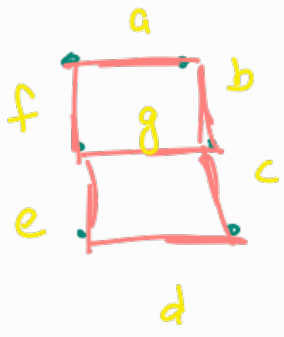
BCD to Seven Segment decoder::



If we want to display 2







with the use of a,b,c,d,e,f,g we can display and.
BCD → inputs meaning 4-inputs
7- outputs.

BCD

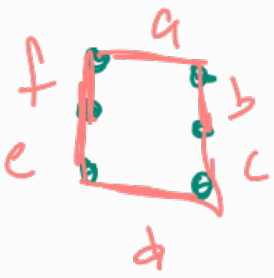
A	B	C	D
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1

Seven Seg

a	b	c	d	e	f	g
1	1	1	1	1	1	0
0	1	1	0	0	0	0
1	1	0	1	1	0	1
1	1	1	1	0	0	1
0	1	1	0	0	1	1
1	0	1	1	0	1	1
1	0	1	1	1	1	1
1	1	1	0	0	0	0
1	1	1	1	1	1	1
1	1	1	1	0	1	1

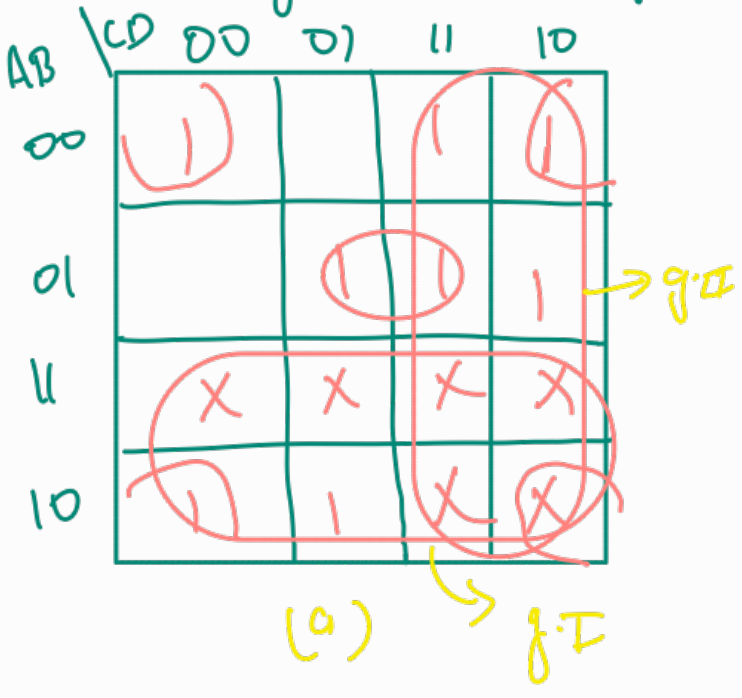
for a when A B C D meaning

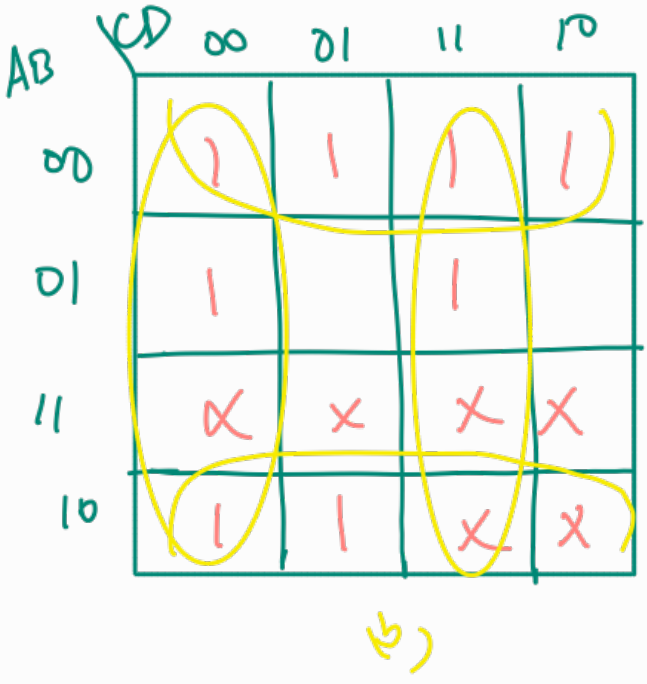
0 0 0 0 display 0



→ must be 1 (except g)

now making 7 K-maps





Encoder example:: 8 bit encoder 8 = 2 inputs

I_7	I_6	I_5	I_4	I_3	I_2	I_1	I_0	MSB A	B	LSB C
0	0	0	0	0	0	0	0	X	X	X
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

output equations:: For A (look where A is 1)

$$A = I_4 + I_5 + I_6 + I_7$$

For $B = I_2 + I_3 + I_6 + I_7$

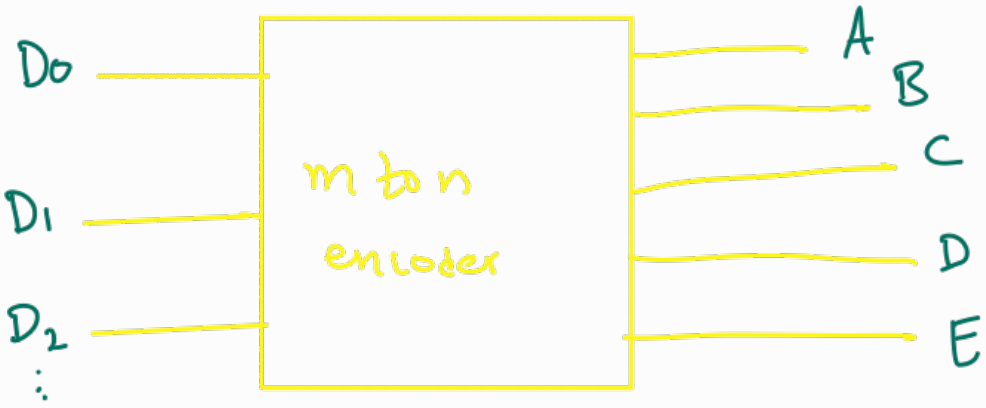
For $C = I_1 + I_3 + I_5 + I_7$

now we can make a circuit diagram

Encoders | opposite of decoders.

flip the inputs | outputs

formula Inputs = 2^{outputs}

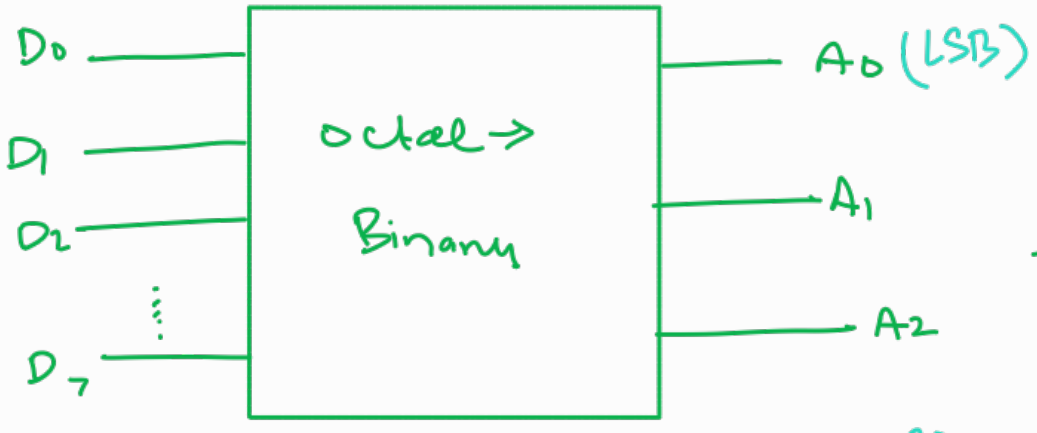


E.g of encoder

→ Octal to Binary encoder

→ Priority encoder.

1) Octal to Binary encoder
(0 - 7)



Output lines

$n = 2^n$

$8 = 2^n$

$8 = 2^3$

3 output

Takes a octal digit and produces a Binary output

(Takes a octal digit and Produces its binary equivalent)

outputs

I_7	I_6	I_5	I_4	I_3	I_2	I_1	I_0	A_2	A_1	A_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

inputs

So when lets say we are looking at D_2 input it means it gives us output that is the decimal equivalent of 2. which is 010 for $A_2\ A_1\ A_0$

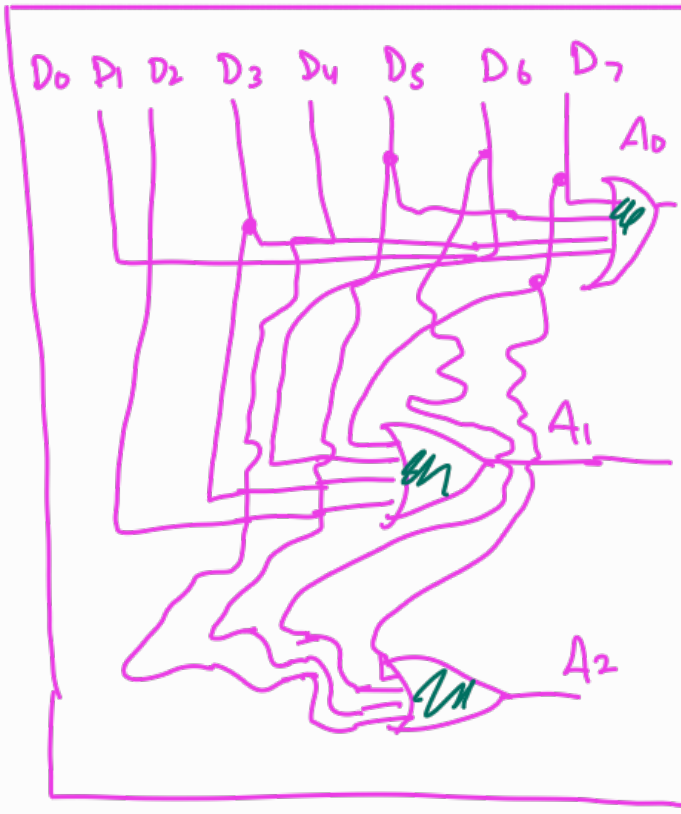
To make equations for $A_2\ A_1\ A_0$. look at where TT value is 1.

I_7	I_6	I_5	I_4	I_3	I_2	I_1	I_0	A_2	A_1	A_0
0	0	0	0	0	0	0	1	0	0	0
0	0	0	0	0	0	1	0	0	0	1
0	0	0	0	0	1	0	0	0	1	0
0	0	0	0	1	0	0	0	0	1	1
0	0	0	1	0	0	0	0	1	0	0
0	0	1	0	0	0	0	0	1	0	1
0	1	0	0	0	0	0	0	1	1	0
1	0	0	0	0	0	0	0	1	1	1

$$A_0 = D_1 + D_3 + D_5 + D_7$$

$$A_1 = D_2 + D_3 + D_6 + D_7$$

$$A_2 = D_4 + D_5 + D_6 + D_7$$



$A_0 = D_1 + D_3 \ldots\ldots$

(limitation of encoders)

→ if two Inputs are simultaneously high the output produces incorrect combination.

→ when are the inputs are D_3 Output will be generated. and $D_0 = 1$ also gives 0 in output ?

000

Solution → **Priority encoder**

if lets say D_3 and D_6 are both 1 at the same time give priority to D_6. hence Priority encoder. hence output would be $(110)_2$

note 001x means $D_3' D_2' D_1' D_0$ ↓ Don't care

4 bit priority encoder.

Just as we have 2 to 4 decoder know we will have 4 to 2 encoder

Inputs				Outputs		
I_3	I_2	I_1	I_0	A_1	A_2	V
0	0	0	0	x	x	0
0	0	0	1	0	0	1
0	0	1	x	0	1	1
0	1	x	x	1	0	1
1	x	x	x	1	1	1



Valid output → zero means all inputs are off.
→ one means any DNF input is On.

Use don't care where now input value doesn't matter. e.g

D_3	D_2	D_1	D_0
0	0	1	x

we used 'x' for D_0 as we know that highest the piority will be given to D_1 so 'x' for D_0

K-maps $2^{inputs} = 2^4 = 16$ cells. There will be two 4 K-maps

For A_2

D_3D_2 \ D_1D_0	00	01	11	10
00	X		1	1
01				
11	1	1	1	1
10	1	1	1	1

D_3	D_2	D_1	D_0	A_1	A_2	V
0	0	0	0	x	x	0
0	0	0	1	0	0	1
0	0	1	x	0	1	1
0	1	x	x	1	0	1
1	x	x	x	1	1	1

D_3	D_2	D_1	D_0
0	0	1	x

can be

0	0	1	1
0	0	1	0

1	x	x	x	
1	0	0	0	
1	0	0	1	
1	0	1	0	
1	0	1	1	
1	1	0	0	
1	1	0	1	
1	1	1	0	
1	1	1	1	

Also mapping A_2 don't care. (X) as it can be 1.

0000

To fill up the k-map look at the outputs and if there is 1 in the output look at the values of corresponding inputs D_3 D_2 D_1 .... D_0. Also include 'x' in k-map if 'x' is present on the output side

equations

$A_2 = D_3 + D_1\overline{D_2}$

$A_1 = D_2 + D_3$

note:- Valid input is just OR of all inputs.

$V = D_3 + D_2 + D_1 + D_0$

Expansion of decoders:-

By the use of enable input which allows use of smaller decoders to build bigger decoders.

now using (2) 2x4 decoders to build 3x8 decoder.

LSB

A_1	A_0
0	0
0	1
1	0
1	1

2 x 4

A_2	A_1	A_0
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

3 x 8

1) Always first make the input combinations of the required decoders.

2) now finding the table of 2x4 in the table of 3x8

3) now start from LSB of 3x8 decoder and map the table of 2x4 decoder on it.

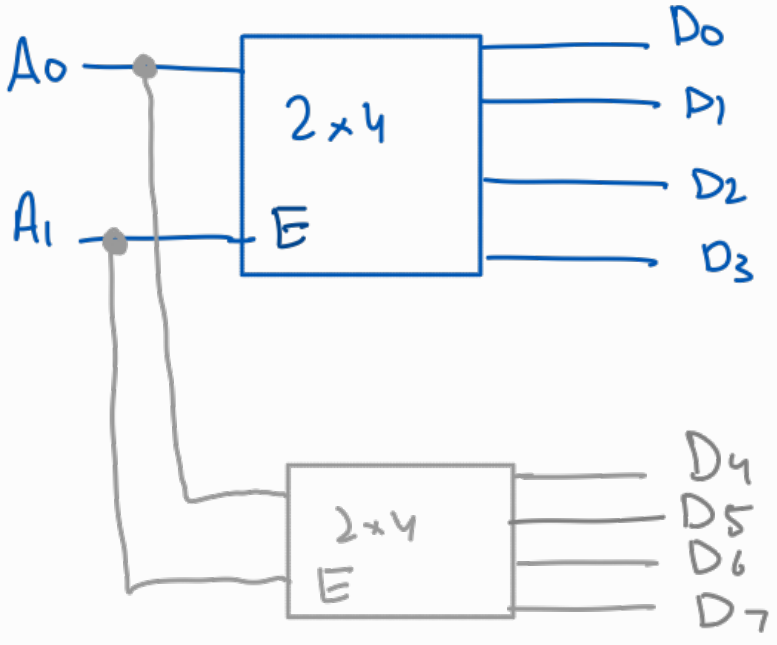
A_1	A_0
0	0
0	1
1	0
1	1

2 x 4

A_2	A_1	A_0
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

3 x 8

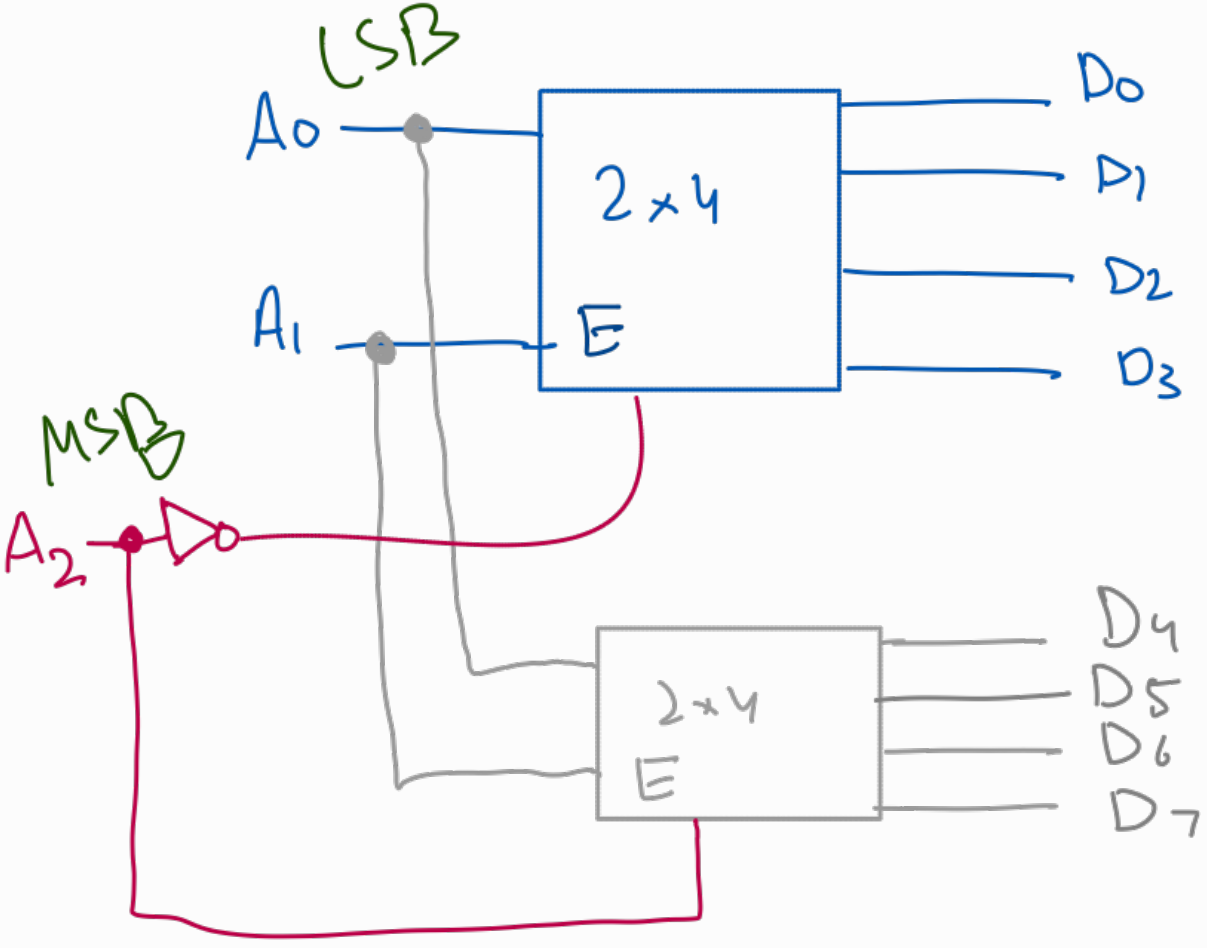
LSB



Making inputs common as 3x8 can have only 3 inputs

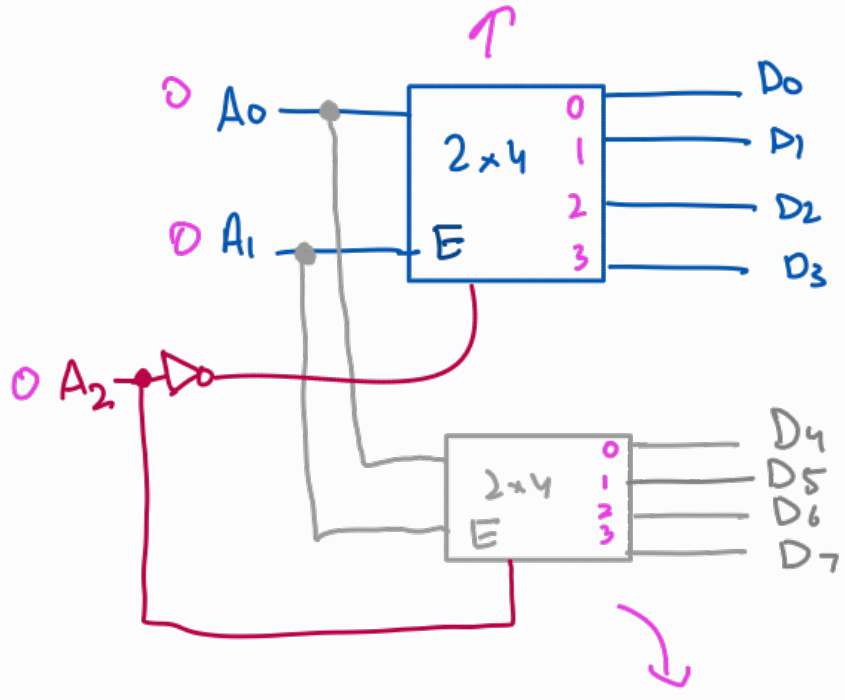
For the MSB we now use the enable pin (E)

one passes through not gate the other doesnot.



0 NOT =1

this is on as will give output for $A_0=0$ and $A_1=0$ $= D_0$



Testing for the first combination

A_2	A_1	A_0
0	0	0

this decoder is off as E=0

Now making 4x16 decoder using 3x8 decoders

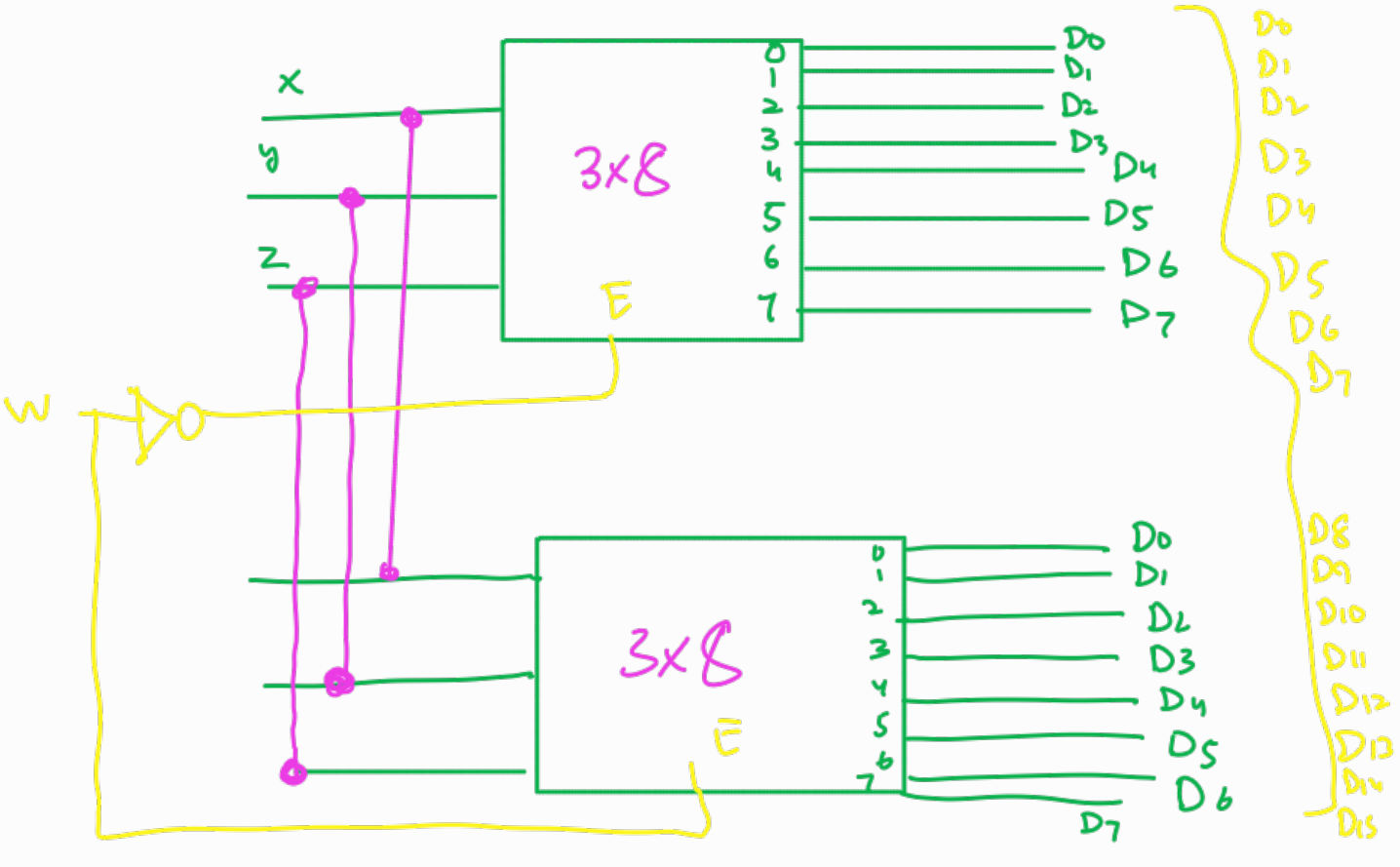
A_2	A_1	A_0
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

3x8

now starting from the LSB of 4x16 decoder to map 3x8 decoder

w	x	y	z
0	0	0	0
0	0	0	1
0	0	1	0
0	0	1	1
0	1	0	0
0	1	0	1
0	1	1	0
0	1	1	1
1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1

4x16



Enable will always be used as MSB

Trick to know how many decoders we need if we are not given its number.

e.g construct a 3 to 8 decoder using 2 to 4 decoder (s)

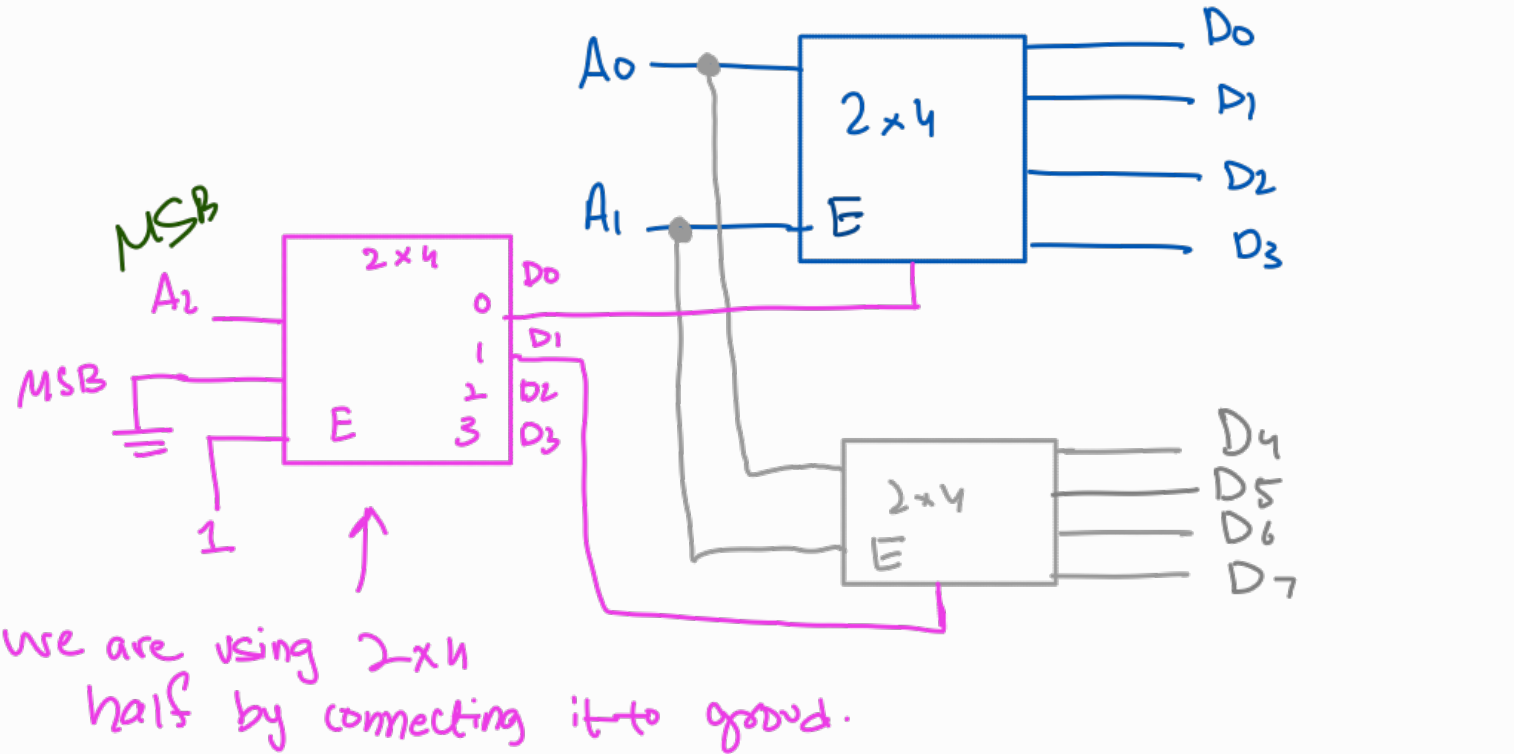
Divide the outputs

$\frac{8}{4} = 2$ → meaning we need 2 → 2×4 decoders

↓

continuous division until the Answer is Smaller or 1 as compared to 4

when we are not given the no of decoders required we can also not use Logic gates for enable input as we need purely 2×4



we are using 2×4 half by connecting it to ground.

ground → permenantly zero

only possible combinations

g	A_2
0	1
0	0

hence D_2 D_3 cannot come.

1	0
1	1

✗ not possible.

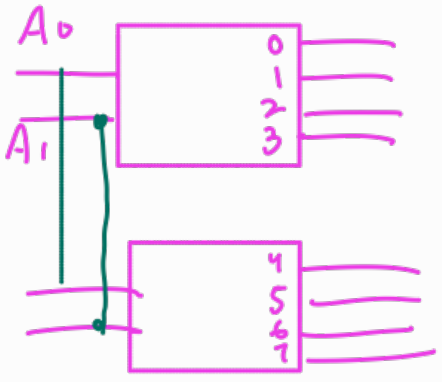
Make 4x16 from 2x4 decoder.

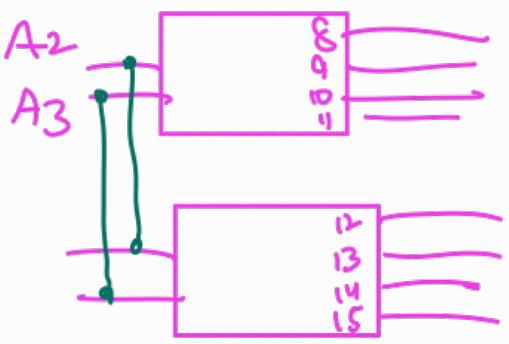
$\frac{16}{4} = 4$ $\frac{4}{4} = (1)$

Stop here

now add 4+1 → 5 2x4 decoder needed.

A_3	A_2	A_1	A_0

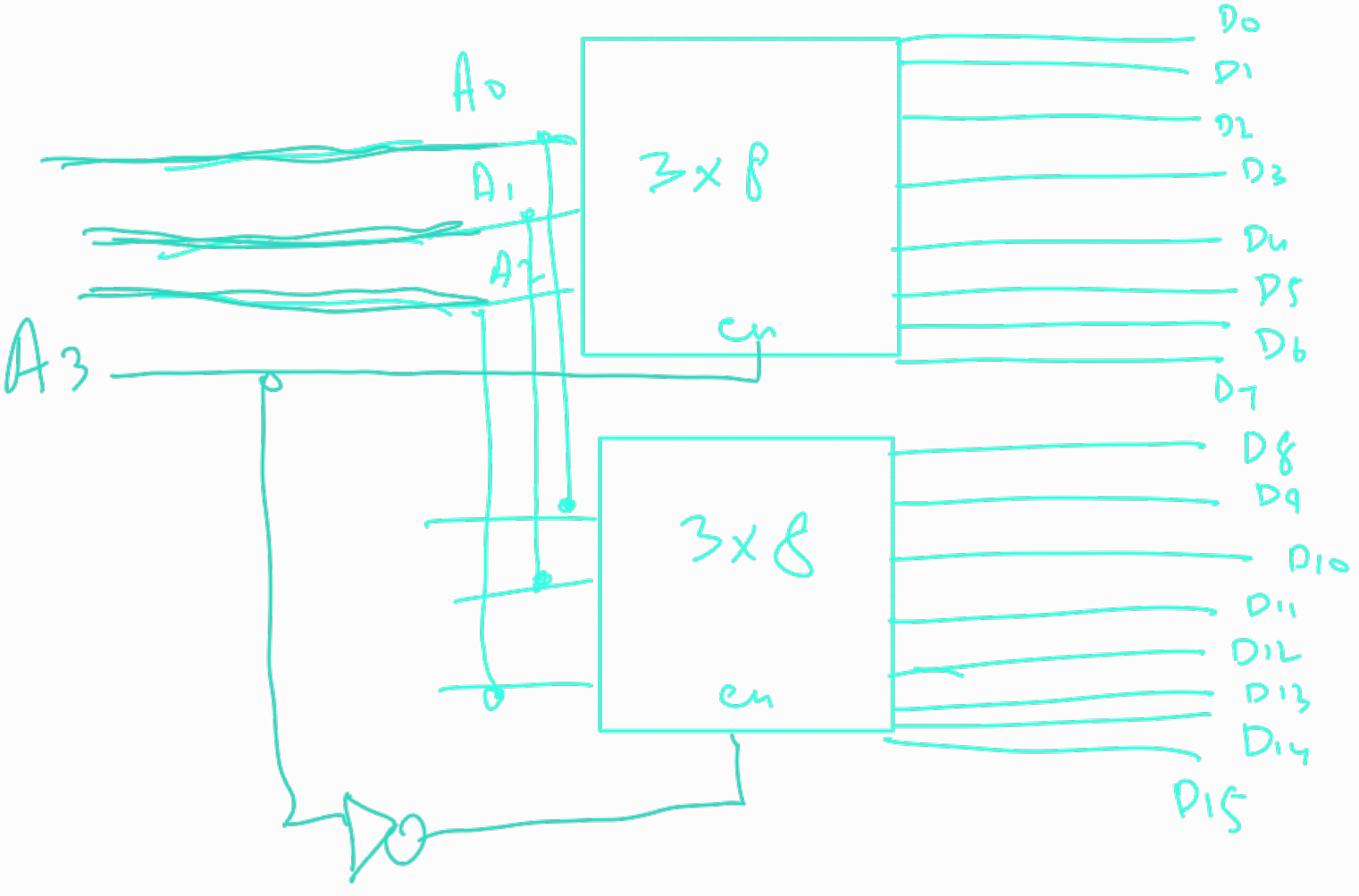




M( ) → POS | Maxterm

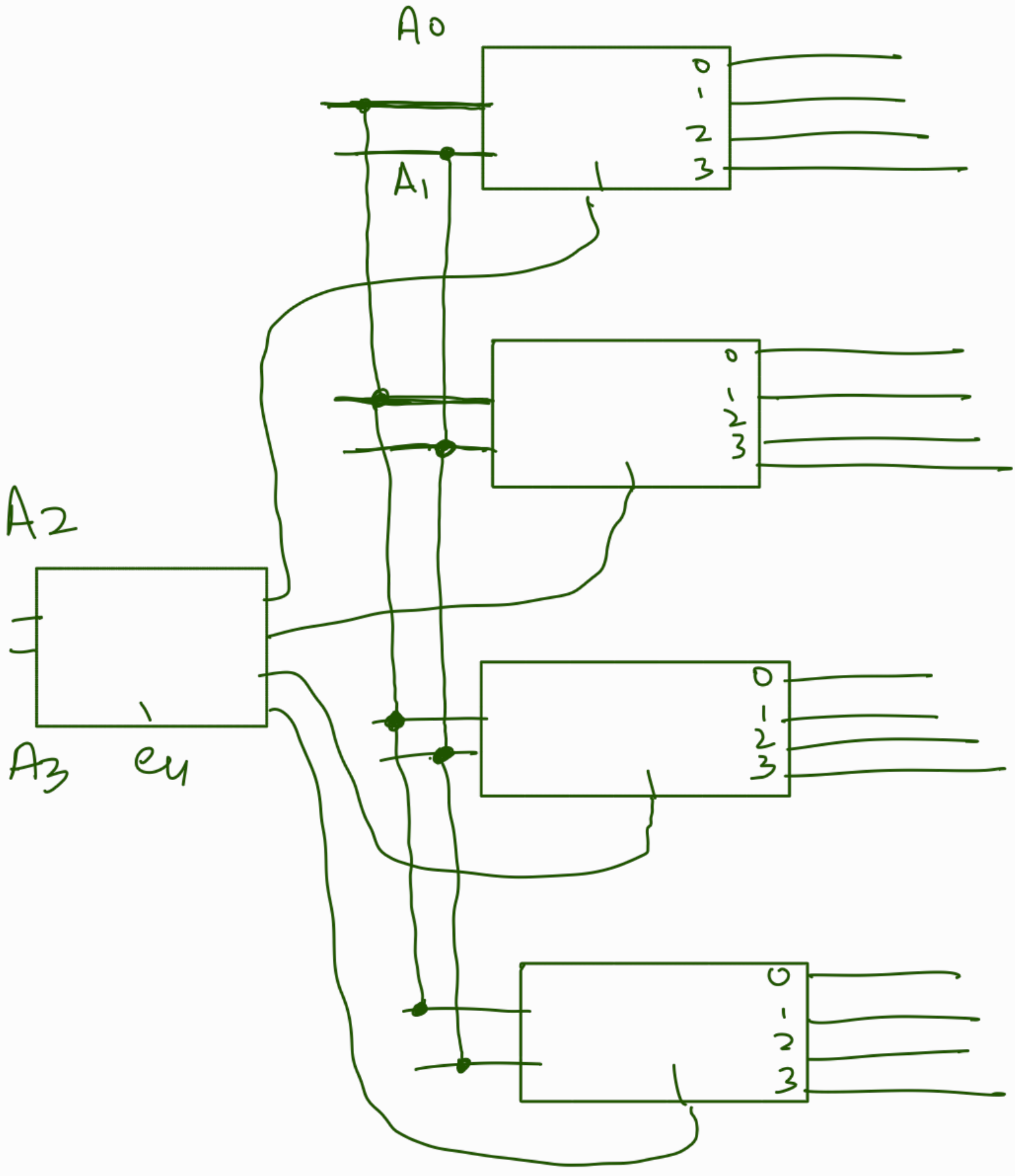
Σ( ) → SOP | Minterms.

(4x16) using (3x8) decoder

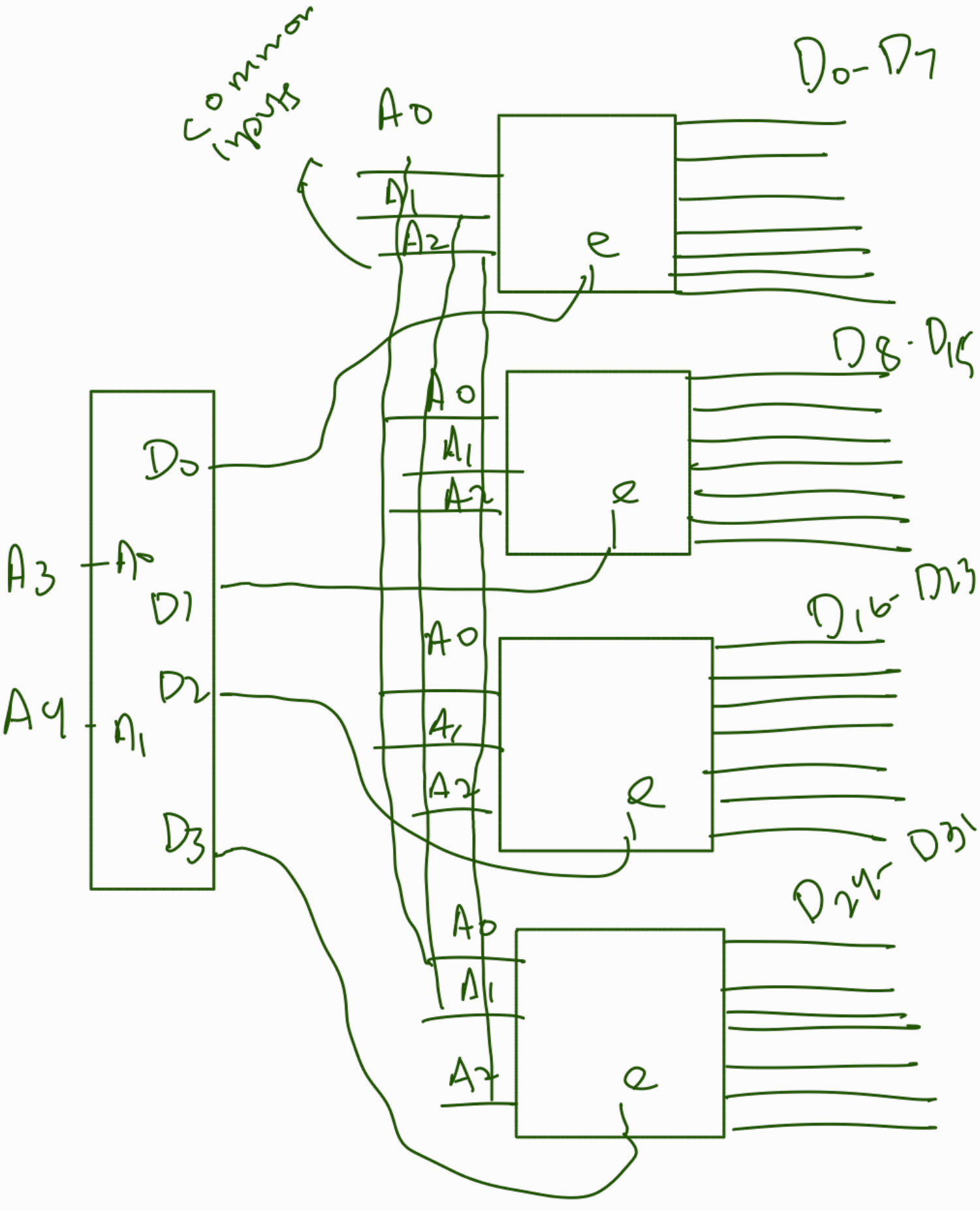


Making 4x16 decoder from 2x4 decoders (5 → 2x4)

(A3 A2 A1 A0)



Constructing 5x32 decoder using 4 (3x8) and one 2x4 Decoder only.



Note: Decoders output is minterms

Q Design the given function using decoder and an OR gate

$$F(x,y,z) = \bar{x}\bar{y}\bar{z} + \bar{x}\bar{y}z + x\bar{y}\bar{z} + xyz$$

$$= 000 + 001 + 110 + 111$$

x	y	z	F	
0	0	0	1	m_0
0	0	1	1	m_1
0	1	0	0	m_2
0	1	1	0	m_3
1	0	0	0	m_4
1	0	1	0	m_5
1	1	0	1	m_6
1	1	1	1	m_7

